International Institute for Special Education (IISE)

College Code: 057

Master of Computer Applications

MCA Second Semester

Session: 2025-2026

Subject: Data Structures & Analysis of Algorithms Lab

Subject Code: BMC254

Submitted To: **Submitted by:**

Mr. Naveen Upreti **(Student Name)**

BMC254: Data Structures & Analysis of Algorithms Lab

CO Code	Course Outcome(CO)	Bloom's Level (KL)
CO1	Write and execute programs to implement various searching and sorting algorithms.	K_3
CO2	Write and execute programs to implement various operations on two-dimensional arrays.	K_3
CO3	Implement various operations of Stacks and Queues using both arrays and linked lists data structures.	K_3
CO4	Implement graph algorithm to solve the problem of minimum spanning tree.	K_3

Practical File Index

S.No	Practical	Course Outcome	Date	Page No.	Signature of faculty
1.	Write a program to implement insertion and deletion in an array.	CO1			
2.	Write a program to implement Linear Search using iteration and recursion through switch case.	CO1			
3.	Write a program to implement Binary Search using iteration and recursion through switch case.	CO1			
4.	Write a program to implement Tower of Hanoi problem.	CO1			
5.	Write a program to implement Bubble and Optimized Bubble sort.	CO1			
6.	Write a program to implement Selection sort.	CO1			
7.	Write a program to implement Insertion sort.	CO1			
8.	Write a program to implement Merge sort.	CO1			
9.	Write a program to implement Quick sort.	CO1			
10.	Write a program to implement Heap sort.	CO1			
11.	Write a program to implement Count sort.	CO1			
12.	Write a program to implement following operations with Matrices: Transpose, Addition, Subtraction and Multiplication.	CO2			
13.	Write a program to implement Sparse Matrix.	CO2			
14.	Write a program to implement Matrix Multiplication by Strassen's algorithm.	CO2			
15.	Write a program to implement a Single Linked list.	CO3			
16.	Write a program to implement a Doubly Linked list.	CO3			
17.	Write a program to implement a Circular Singly Linked list.	CO3			

18.	Write a program to implement a Circular Doubly Linked list.	CO3			
19.	Write a program to implement Stack using array.	CO3			
20.	Write a program to implement Stack using linked list.	CO3			
21.	Write a program to implement Linear queue using array.	CO3			
22.	Write a program to implement Linear queue using linked list.	CO3			
23.	Write a program to implement Circular queue using array.	CO3			
24.	Write a program to implement Circular queue using linked list.	CO3			
25.	Write a program to evaluate a Postfix expression.	CO3			
26.	Write a program to convert an Infix expression to Prefix expression.	CO3			
27.	Write a program to convert an Infix expression to Postfix expression.	CO3			
28.	Write a program to traverse a binary tree in preorder, inorder and postorder.	CO4			
29.	Write a program to implement Matrix Multiplication by Strassen's algorithm.	CO4			
30.	Write a program to implement Kruskal's algorithm of Minimum Spanning Tree(MST)	CO4			
31.	Write a program to implement Prim's algorithm of Minimum Spanning Tree(MST)	CO4			